Rethinking Generalization in Reasoning SFT: A Conditional Analysis on Optimization, Data, and Model Capability

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Research Questions

- Under what conditions does supervised fine-tuning (SFT) with long chain-of-thought supervision generalize across domains in reasoning tasks?
- How do optimization dynamics, training data quality/structure, and base model capability jointly affect cross-domain generalization in reasoning SFT?
- What are the trade-offs between reasoning performance gains and safety degradation in long-CoT SFT?

Core Idea

This paper challenges the prevailing "SFT memorizes, RL generalizes" narrative by showing that reasoning SFT's cross-domain generalization is conditional rather than absent. The key insight is that apparent generalization failures are often artifacts of experimental conditions - insufficient training (dip-and-recovery dynamics), poor data quality, or weak base models. The core trick is systematic ablation across three factors: optimization sufficiency (training long enough to reach recovery phase), data structure/quality (verified long-CoT traces vs low-quality solutions), and model capability (stronger models internalize transferable procedural patterns while weaker ones imitate surface verbosity).

Key Figure

Preprint. Under review.

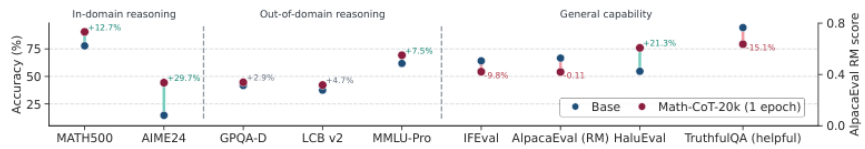

Figure 2: Replication of previous findings: short-epoch reasoning SFT improves math substantially but yields limited or negative gains on OOD reasoning and general capabilities.

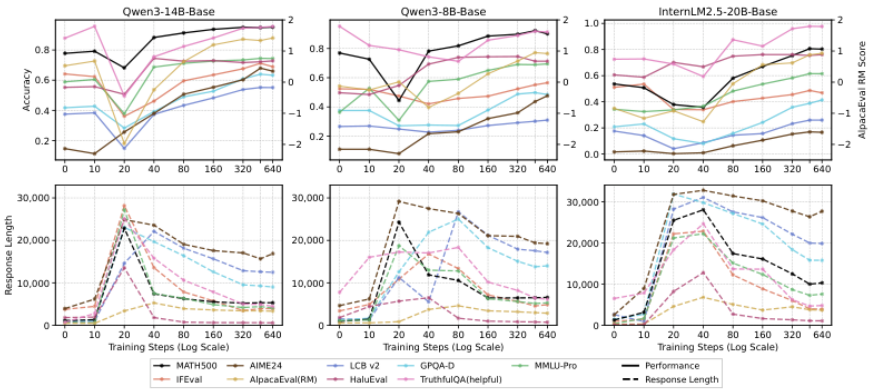

Figure 3: Training dynamics of long-CoT reasoning SFT. **Top:** benchmark performance over training steps, showing a common dip-and-recovery pattern on most benchmarks. **Bottom:** response length over training steps, with an early length surge followed by a gradual pullback to shorter responses. Results on Qwen2.5 models are provided in App. C.2.

Figure 3: Shows the critical dip-and-recovery pattern across multiple models and benchmarks, with performance initially degrading before recovering and improving, demonstrating that early checkpoints systematically underestimate SFT generalization potential.

Limitations

- Experiments limited to math reasoning domain - generalization of findings to other reasoning domains (code generation, scientific reasoning, multi-modal tasks) remains unclear
- Computational constraints restrict experiments to dense models up to 20B parameters - scalability to larger models or mixture-of-experts architectures not validated
- No direct comparison with RL-based methods, making it difficult to establish relative advantages under different experimental conditions
- Safety degradation appears inherent to long-CoT procedural patterns with no clear mitigation strategy proposed